

AODN Cloud Optimised Library

From 20 years of IMOS NetCDF to cloud-ready analysis

Laurent (Loz) Besnard - AODN Data Scientist

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The Data Access bottlenecks

Over two decades of oceanographic data needed to transition from file-based downloads to direct cloud querying.

User Friction:

- **Download:** data had to be retrieved as physical NetCDF files from the [AODN Portal](#) or [THREDDS](#)
- **Opening files:** analysing a dataset often meant downloading and concatenating hundreds of NetCDF files
- **Simple questions were complicated:** e.g. *“Give me the Sea Surface Temperature time series at this location over 20 years”* could require downloading gigabytes of data

AODN overhead:

- **Database:** NetCDF files were ingested into PostgreSQL and exposed through GeoServer/PostGIS
- **Infrastructure to maintain:** database instances, VMs, replication, backups, etc.
- **Not directly accessible:** although the data was queryable through the database, the database itself was not publicly available to users

The Solution: Cloud Optimised (CO) Formats

Data type	Format
Tabular / time series	Parquet
Gridded / multidimensional	Zarr

Zarr Chunk

MULTIDIMENSIONAL



What is a Chunk? A sub-block of an N-D array grid (e.g. Time $\times$ Lat $\times$ Lon).

Cloud Superpower: HTTP Byte-Range requests fetch *only* the needed 3D chunks without loading the entire multi-terabyte dataset.

Parquet Partition

TABULAR / ROWS

 dataset / year=2026 /

 month=01.parquet

 month=02.parquet

 dataset / year=2025 / ...

What is a Partition? Splitting table rows into subdirectories by columns (e.g. Year/Month).

Cloud Superpower: "Partition Pruning" skips irrelevant folders completely, reading only targeted files during SQL queries.

Selective I/O

Instead of downloading full 5GB files, cloud clients request precise byte offsets for specific chunks or files over HTTP S3 connections.

Parallel Reading

Tools like Dask, DuckDB, and Polars can fetch hundreds of chunks simultaneously in parallel threads across distributed clusters.

We created the [AODN Cloud Optimised library](#) to transform the existing NetCDF files into **open, analysis-ready** formats:

Built on open-source libraries:

NetCDF4	xarray	Zarr	Dask	Apache Parquet	s3fs
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The resulting is:

- over 150 CO datasets (Timeseries, Trajectories, TrajectoryProfiles, Profiles, Gridded ...)
- stored in a **public AWS S3 bucket** `s3://aodn-cloud-optimised`
- through the [AWS Open Data](#) program, making them publicly accessible.

Bonus: the library is not limited to IMOS only data.

We partnered with **CSIRO** to use the library to convert their underway data to Parquet, making it possible to bring their data into the AODN ecosystem and improve interoperability between datasets.

The Processing Challenges

Heterogeneous input	Dynamic datasets	Processing at scale
NetCDF3/4 / CSV Different formats	Datasets are not static Append new data - Update existing data - Remove / replace data - Metadata can change	Thousands of files Initially: Coiled + Dask Today: Prefect Cloud + Dask

No satisfying tool existed for our requirements.

The Technology is still relatively new and mostly used for static datasets, with no widely adopted conventions on how to format the data.

Generic Processing

The library is designed to handle as many datasets as possible through:

- common **Zarr and Parquet handlers**
- **configuration-driven** processing without changing the processing code for 99% of datasets

Some datasets, however:

- have non-conventional structures (e.g. Argo)
- are multi-multidimensional datasets, but fit better as tabular representations (Bio-Optical Database)

For these datasets, a **custom handler** can be created to extract and transform (such as flattening) the data before writing it to the cloud-optimised format.

Configuration-Driven Processing

Everything about a dataset is defined in one configuration file:

- input data location
- schema definition (variables, dimensions, variable attributes ...)
- schema transformation (e.g. creating new variables from global attributes)
- chunking and partition strategy
- AWS OpenData metadata
- cluster processing settings

```
{
  "dataset_name": "vessel_fishsoop_realtime_qc",
  "cloud_optimised_format": "parquet",

  "schema": {
    "TRAJECTORY_ID": { "type": "double", "long_name": "..." },
    "TIME": { "type": "timestamp[ns]", "standard_name": "time" },
    "LATITUDE": { "type": "float", "valid_min": -90, "valid_max": 90 },
    "TEMPERATURE": { "type": "float", "units": "degrees_Celsius" }
  },

  "schema_transformation": {
    "partitioning": [
      { "type": "time_extent", "partition_period": "M" },
      { "type": "spatial_extent", "spatial_resolution": 5 }
    ]
  },

  "run_settings": { "cluster_mode": "coiled", "batch_size": 300 }
}
```

No code changes. Configuration inheritance. Reproducible.

How to Access

Self-contained Jupyter Notebooks

Each cloud-optimised dataset has a dedicated [example notebook](#):

- Self-contained examples
- Designed to run in **Google Colab**
- No need to download the full dataset

GetAodn Python Class

The notebooks use a custom Python class to abstract:

- Zarr / Parquet access
- Dataset partitioning
- Data queries

```
from DataQuery import GetAodn

aodn = GetAodn()

dataset_name = "diver_cryptobenthic_fish_abundance_qc"
aodn_dataset = aodn.get_dataset(dname)

df = aodn_dataset.get_data(date_start='2015-12-01',
                           date_end='2019-01-01',
                           lat_min=-34,
                           lat_max=-31,
                           lon_min=151,
                           lon_max=153,
                           )
```

Notebooks Presentation

Vessel FishSOOP

Wave Buoy Realtime

Satellite SST L4 RAMSSA

Mooring Hourly Timeseries

Argo

Thank you

Laurent.Besnard@utas.edu.au